Research proposal

Temporal Client Reputation for Byzantine-Resilient Federated Intrusion Detection

Mashrur Faiyaz, Ahnaf Shahriar Pias
Software Engineering, Islamic University of Technology
October 2026

Working draft based on the consulted papers and recorded baseline experiments. The reputation mechanism and experimental extensions below remain proposed work.

1 Project summary

Federated learning allows organisations to train a shared intrusion detection model while keeping their network traffic data locally. Its usefulness depends on the quality of the updates that participating organisations submit. A compromised participant can send misleading updates, while an honest participant may produce an unusual update because its traffic differs from that of other participants. A defence must distinguish these cases without excluding useful information about uncommon attacks.

This project will investigate a temporal client reputation mechanism for federated intrusion detection. Its reputation foundation is AntiFLipper by Aashnan Rahman and co-authors, while its CICIoT2023 motivation comes from the federated intrusion detection study by Arivukarasi and co-authors. The central question is whether adapting this foundation to heterogeneous network traffic improves resistance to malicious updates while preserving honest-client contributions.

Current experiments on CICIoT2023 provide centralized and federated neural-network baselines. The next stage will extend these experiments to controlled poisoning scenarios, longer training runs, and repeated trials. The intended outputs are an evaluated reputation mechanism, a reproducible experimental pipeline, and evidence about its effectiveness, cost, and limitations.

2 Introduction

2.1 Problem and motivation

Network intrusion detection classifies traffic as benign or as a type of attack. In a cross-silo federated setting, each client represents an organisation or network domain that trains a local model. The server combines client updates into a shared model. Differences in traffic composition create non-independent and identically distributed (non-IID) client data, including unequal attack frequencies and uneven representation of rare classes.

Byzantine clients can submit arbitrary updates or manipulate local training data. Their updates may reduce overall detection quality or selectively hide an attack class. Honest non-IID clients can also disagree with the majority. Treating disagreement alone as evidence of malicious behaviour therefore risks suppressing legitimate contributions. This project will test whether evidence accumulated across rounds can support better aggregation decisions.

Two consulted studies define the project's starting point. AntiFLipper supplies the trust-based defence foundation [1]. Arivukarasi et al.'s CICIoT2023 study motivates the application setting

and explicitly identifies adversarial aggregation and client reputation as directions for further work [2]. The proposed contribution will be assessed as an adaptation and evaluation in this setting.

2.2 Research questions

RQ1. Does temporal reputation improve macro-F1 and attack-class recall under poisoning compared with aggregation without reputation?

RQ2. How often does the mechanism downweight or flag honest non-IID clients, especially clients containing rare attack classes?

RQ3. How does it respond to persistent, intermittent, and delayed malicious behaviour, and can an honest client recover from an adverse assessment?

RQ4. What computation, communication, and validation-data costs accompany any improvement?

2.3 Research approach and significance

The study will use controlled simulations in which data partitions, malicious client identities, and attack schedules are known. It will compare aggregation rules under matched training conditions and examine both global detection performance and client-level treatment. The working hypothesis is that a carefully calibrated history of contribution quality can reduce poisoning damage without systematically disadvantaging honest clients whose traffic differs from the majority. This hypothesis remains to be tested.

The practical value lies in establishing when temporal evidence helps and when it fails. A successful outcome would support intrusion detection across organisations that cannot pool their traffic data. An unsuccessful outcome would identify failure conditions, such as unreliable validation coverage or an attacker that first builds a favourable history.

2.4 Objectives

1. Establish reproducible centralized and federated baselines with class-sensitive evaluation.
2. Design a reputation update and aggregation rule that considers contribution history and honest data heterogeneity.
3. Evaluate poisoning resistance, honest-client treatment, and recovery across controlled non-IID conditions.
4. Measure the contribution of temporal memory through ablations and report the computational cost and scope of the findings.

3 Literature review

3.1 Reputation foundation: AntiFLipper

Rahman et al. introduce AntiFLipper as a defence against label flipping [1]. Clients evaluate the global model on their local training data and report accuracy. The server adjusts trust using signed squared deviations from the participating clients' mean accuracy. It accumulates flags when trust falls below a threshold, excludes repeatedly flagged clients, and weights retained models by trust. Its evaluation uses MNIST and CIFAR-10 under IID and non-IID partitions and includes delayed and intermittent attacks.

Trust accumulation and changing attack schedules are therefore established parts of this foundation. This project will examine its transfer to intrusion detection, the treatment of rare-class clients, and the reliability of reported scores under model manipulation.

3.2 CICIoT2023 motivation and the security limitation

Arivukarasi et al. propose a CICIoT2023 framework combining federated autoencoder learning, LBIID feature selection, and a HiViT-IDS classifier, with FedProx used to address client drift [2]. This is the consulted study that motivates the project's dataset choice. Section 4.3 acknowledges an assumption of honest participants and a lack of explicit defence against data and model poisoning. It identifies trust-weighted updates and reputation scoring as possible extensions.

This motivates testing a defence under malicious-client participation. The study's classification figures will not be used as expected results for the current MLP: the architecture, preprocessing, and training protocol differ. The proposed evaluation will establish its own clean and attacked baselines under matched conditions.

3.3 Related aggregation and evaluation work

Kritharakis et al. evaluate submitted models on trusted server data, cluster their losses into low- and high-loss groups, and aggregate the low-loss group [3]. Their reported image-classification experiments use IID partitions and include label flipping, sign flipping, and Gaussian noise. This provides a relevant comparison for server-observed scoring, with the trusted-data assumption stated explicitly.

Jimenez G. et al. distinguish forms of non-IID data and review partition protocols and heterogeneity measures [4]. This informs the decision to document label skew, sample-count imbalance, and class coverage rather than describe a partition only as non-IID. ENIDS combines CNN, LSTM, and GRU base learners with a DNN meta-learner and addresses feature selection and imbalance on UNSW-15 and CICIDS-2017 [5]. It informs classifier and preprocessing choices; its ensemble is outside the initial aggregation study.

Gourceyraud et al. investigate federated unsupervised intrusion detection using clustering [6]. Their study provides context for collaborative detection with limited labels. Zhao et al.'s privacy survey explains how model updates can expose information about local data [7]. The present study consequently treats poisoning resistance and privacy as separate properties.

3.4 Research focus

The consulted work supports a focused question: how does a trust-based defence behave when honest intrusion-detection clients have strongly unequal class distributions and malicious clients can alter updates or reported scores? The planned comparison will test AntiFLipper alongside a proposed adaptation on CICIoT2023. It will examine minority-class recall, honest-client downweighting, and recovery as well as global performance. This is an empirical extension of the identified foundation. Any broader novelty claim will require a wider literature check.

4 Current progress

The available experiment summaries record 36,014,594 training samples and 9,003,649 test samples, consistent with an 80/20 split. Each experiment used 39 input features, 34 classes, and a multilayer perceptron (MLP) with layer sizes 39–128–64–34. The comparison records

seed 42, batch size 8,192, learning rate 0.001, and three effective data passes. The repository notebooks configure ten clients and Dirichlet label partitioning with $\alpha = 0.1$ for the non-IID runs. FedAvg and FedProx used the same non-IID client partition. Each federated run contains three communication rounds.

Table 1: Preliminary results from the recorded baseline runs. Accuracy and balanced accuracy are percentages; macro-F1 is on a 0–1 scale.

Experiment	Accuracy (%)	Balanced acc. (%)	Macro-F1
Centralized MLP	77.93	55.37	0.5670
FedAvg, IID	77.16	52.57	0.5290
FedAvg, non-IID	65.74	36.24	0.2880
FedProx, non-IID	64.91	31.28	0.2336

These runs show lower performance under the recorded non-IID partition, particularly when classes receive equal weight in macro-F1. They motivate closer examination of client heterogeneity and minority-class detection. Three rounds and one recorded seed do not establish convergence or a general ranking of the methods. The available summaries do not establish Byzantine resistance, uncertainty across repeated trials, or performance of the proposed reputation mechanism.

A centralized XGBoost comparison using the project's own dataset and an 80/20 split is a planned diagnostic extension. Its results must be verified before inclusion. The MLP remains the initial federated model because its shared parameter structure supports the proposed update aggregation. XGBoost will serve as a centralized comparison, with its training and evaluation conditions reported separately.

5 Methodology

5.1 Data preparation and evaluation split

CICIoT2023 will be the primary dataset, following the application setting of Arivukarasi et al. [2]. The pipeline will record source files, retained rows, label mappings, missing-value handling, duplicate handling, and feature transformations. The recorded experiment counts will be reconciled with the finalized preprocessing pipeline before further comparisons.

The primary evaluation will retain an 80/20 training–test allocation. A validation subset will be drawn only from the training portion for model selection and reputation calibration. Transformations will be fitted on the local or designated training data according to the stated protocol. Test data will not inform tuning, reputation scores, or stopping decisions. Where reliable capture, device, or session identifiers exist, a grouped split will provide a supplementary evaluation of generalisation beyond randomly separated records. Any cross-split duplication will be audited and its treatment documented. CICIDS2017 may provide a secondary validation dataset if time and resources permit.

5.2 Federated setting and baselines

The initial design will simulate 10–20 cross-silo clients with a shared MLP architecture. IID partitions will provide a reference condition. Non-IID partitions will vary label skew and client sample counts, with partition parameters and class coverage recorded. Synthetic partitions will be described as simulations rather than evidence of actual organisational traffic distributions.

Comparisons will include FedAvg, FedProx, and a reproduction of AntiFLipper's scoring, flagging, and trust weighting adapted to the shared MLP. The loss-based clustering method is a candidate comparison when trusted server data are available. Coordinate-wise median and trimmed mean will provide additional controls if resources permit. Methods will share client partitions, attack schedules, and model architecture where applicable. Local training effort, communication rounds, and tuning budgets will be recorded explicitly. Every adaptation will be documented.

5.3 Threat model

The initial setting assumes an honest server, stable authenticated client identities, and a majority of honest clients. Following the repository's stated scope, experiments will cover label flipping, sign flipping, and random updates. Label flipping will include a targeted mapping that reduces detection of a selected attack class. Initial malicious-client fractions will be 0%, 10%, 20%, and 30%, with the realized client counts reported for each configuration.

Persistent, intermittent, and delayed attack schedules will test transfer to the intrusion-detection setting. A further stress test will allow independent attackers to misreport evaluation scores while manipulating updates. Malicious servers, compromised IoT hardware, privacy inference, communication interception, backdoors, and colluding Byzantine clients remain outside the repository's current scope. Sybil identities and formal security or privacy guarantees are also outside the initial evaluation.

5.4 Candidate temporal reputation mechanism

The design will begin by reproducing the identified reputation foundation. Proposed modifications will test class-sensitive contribution scoring, bounded temporal memory, and reversible downweighting. The equations below specify an initial candidate for these modifications; its parameters and scoring rule remain to be validated.

For each participating client i at round t, the server will compute a normalized contribution score $q_i^{(t)} \in [0, 1]$. Candidate evidence includes changes in class-balanced validation performance and bounded update diagnostics. An initial temporal update is

$$r_i^{(t)} = \beta r_i^{(t-1)} + (1 - \beta) q_i^{(t)}, \qquad 0 \leq \beta < 1, \tag{1}$$

where $r_i^{(t)}$ denotes reputation and β controls memory. Participating clients will begin with the same reputation. One candidate aggregation rule is

$$w_i^{(t)} = \frac{n_i r_i^{(t)}}{\sum_{j \in \mathcal{S}_t} n_j r_j^{(t)}}, \qquad \theta^{(t+1)} = \theta^{(t)} + \sum_{i \in \mathcal{S}_t} w_i^{(t)} \Delta_i^{(t)}, \tag{2}$$

where n_i is the verified training sample count, $\mathcal{S}_t$ is the participating client set, and $\Delta_i^{(t)}$ is the client update. A zero denominator will trigger rejection of that round's updates and retention of the previous global model. Handling of absent clients, weight caps, and any exclusion threshold will be specified before evaluation.

The contribution score is the main open design question. Client reports will be treated as untrusted when arbitrary update manipulation is permitted. A server-observed variant will score submitted models on a small trusted validation set drawn only from the training allocation. This follows the scoring location used in loss-based clustering [3]; the proposed temporal weighting is a separate design choice. Experiments will vary validation size and class coverage. Recovery

rules and weight caps will be examined for their effects on honest clients and on attackers that accumulate reputation before poisoning.

5.5 Evaluation and ablations

Macro-F1 and per-class recall will be the primary detection measures. Accuracy, balanced accuracy, weighted F1, Matthews correlation coefficient, and confusion matrices will provide supporting evidence. Targeted attacks will be assessed by target-class recall and a precisely defined attack-success measure. A binary benign-versus-attack view will report benign false-positive rate and attack detection rate.

Client-level evaluation will measure honest-client flagging where thresholds are used, aggregation weight assigned to honest clients, poisoning detection delay, and recovery time. Results will be broken down by client class coverage and sample count. End-to-end runtime, server scoring cost, peak memory, and communicated bytes will quantify overhead. Simulated runtime will not be presented as real deployment latency.

At least three seeds will be used for the main comparisons, pairing partitions and attack schedules across methods. Means and variation across runs will be reported. Round budgets will be chosen from validation learning curves and available compute, then fixed for matched comparisons. The existing three-round runs will serve as preliminary baselines. Ablations will compare no reputation, current-round scoring without memory, temporal scoring, and variants of the recovery and contribution-scoring rules.

5.6 Tools, feasibility, and limitations

Experiments will use Python, a neural-network framework, standard classification evaluation tools, and version-controlled configurations. Existing Kaggle-based work provides an initial execution environment. Development will use a fixed, documented subset before selected configurations are evaluated at larger scale. Hardware, software versions, sampling rules, and random seeds will accompany the results.

The main risks are insufficient minority-class coverage, dependence on trusted validation data, and compute costs at the recorded dataset scale. Rare classes will be inspected individually, validation coverage will be varied, and larger runs will be staged after smaller-scale checks. Keeping traffic local does not by itself establish privacy of shared updates. Findings will be limited to the evaluated partitions, attacks, and assumptions.

6 Expected contributions

The intended contribution is an empirical evaluation of a reputation defence and its proposed adaptation to non-IID federated intrusion detection. It will establish whether changes to scoring, memory, and recovery improve poisoning resistance while preserving honest-client contributions. Matched comparisons will identify both useful changes and failure conditions. A reproducible pipeline will record preprocessing, partitions, attacks, seeds, and metrics. Claims of improvement will depend on the completed experiments.

Bibliography

[1] Aashnan Rahman, Abid Hasan, Sherajul Arifin, Faisal Haque Bappy, Tahrim Hossain, Tariqul Islam, Abu Raihan Mostofa Kamal, and Md. Azam Hossain. AntiFLipper: A secure and

efficient defense against label-flipping attacks in federated learning. arXiv:2509.22873v2, 2025. URL https://arxiv.org/abs/2509.22873v2.

[2] M. Arivukarasi, S. Harihara Gopalan, S. Gnanamurugan, R. Dhanapal, A. Ramachandran, and A. Manikandan. Federated autoencoder IDS for IoT: A Fed-ANIDS approach using CICIoT2023. *Systems and Soft Computing*, 8:200482, 2026. doi: 10.1016/j.sasc.2026.200482. URL https://doi.org/10.1016/j.sasc.2026.200482.

[3] Emmanouil Kritharakis, Dusan Jakovetic, Antonios Makris, and Konstantinos Tserpes. Robust federated learning under adversarial attacks via loss-based client clustering. arXiv:2508.12672v1, 2025. URL https://arxiv.org/abs/2508.12672v1.

[4] Daniel M. Jimenez G., David Solans, Mikko A. Heikkilä, Andrea Vitaletti, Nicolas Kourtellis, Aris Anagnostopoulos, and Ioannis Chatzigiannakis. Non-IID data in federated learning: A survey with taxonomy, metrics, methods, frameworks and future directions. arXiv:2411.12377v2, 2024. URL https://arxiv.org/abs/2411.12377v2.

[5] Ibrahim Mohammed Sayem, Moinul Islam Sayed, Sajal Saha, and Anwar Haque. ENIDS: A deep learning-based ensemble framework for network intrusion detection systems. *IEEE Transactions on Network and Service Management*, 2024. doi: 10.1109/TNSM.2024.3414305. URL https://doi.org/10.1109/TNSM.2024.3414305.

[6] Maxime Gourceyraud, Rim Ben Salem, Christopher Neal, Frédéric Cuppens, and Nora Boulahia Cuppens. Federated intrusion detection system based on unsupervised machine learning. arXiv:2503.22065v1, 2025. URL https://arxiv.org/abs/2503.22065v1.

[7] Joshua C. Zhao, Saurabh Bagchi, Salman Avestimehr, Kevin S. Chan, Somali Chaterji, Dimitrios Dimitriadis, Jiacheng Li, Ninghui Li, Arash Nourian, and Holger R. Roth. The federation strikes back: A survey of federated learning privacy attacks, defenses, applications, and policy landscape. arXiv:2405.03636v3, 2025. URL https://arxiv.org/abs/2405.03636v3.